

Project Plan

**From Research Data to Evidence-Based
Recommendations:
Designing and Evaluating an AI-Supported Reporting System for
InnoBeweegLab**

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Group project case: InnoBeweegLab — individual research trajectory

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Revision History

Date	Version	Description
14-09-2026	0.1	Initial draft of the project plan

Abstract

InnoBeweegLab designs active-friendly public spaces and translates resident participation, expert assessment, and quantitative research data into evidence-based recommendations for municipalities. Producing those research reports is currently a slow, manual synthesis task that competes directly with the time InnoBeweegLab would rather spend on fieldwork and community engagement. Concretely, this means investigating how a provider-agnostic, multi-agent reporting architecture can be designed so that heterogeneous qualitative and quantitative research data is reliably and traceably transformed into evidence-based draft reports, while the researcher stays in control of interpretation, conclusions, and recommendations

Glossary

Term	Definition
HBO-i	The shared domain description used by Dutch universities of applied sciences for ICT degrees; defines competency layers (e.g. Software, Interaction) and levels (1-4) used to assess this project.
DOT framework	Design & development-Oriented research Template/Triangulation — Fontys ICT's applied-research method, structured around five research types: Library, Field, Lab, Workshop, and Showroom.
Central design question	The single overarching question this individual trajectory investigates; each DOT research type contributes a sub-question that feeds into answering it.
MAS	Multi-Agent System — an architecture in which multiple specialised LLM-driven agents (e.g. extraction, retrieval, writing) collaborate on a task rather than one model handling everything.
RAG	Retrieval-Augmented Generation — grounding a language model's output in retrieved source content (here: InnoBeweegLab's own research data) rather than relying on the model's internal knowledge alone.
LLM	Large Language Model — the class of AI model (e.g. GPT, Claude, Llama, Phi) used to process and generate text.
Provider-agnostic architecture	A system design that does not hard-depend on one specific AI vendor or model, so the underlying model can be swapped (e.g. local small model vs. hosted API) without redesigning the system.
HITL	Human-in-the-loop — a design pattern where a human (the researcher) reviews, corrects, or approves AI-generated content before it is used, rather than the system acting autonomously.
RAGAS	An automated evaluation framework for retrieval-augmented generation, scoring outputs on dimensions such as faithfulness and factual correctness; used here as inspiration for the Lab-phase evaluation method.
Quantization	A technique that compresses a language model's numerical weights so it needs less memory and can run on more modest hardware, at some cost to output quality.
Prototype	A functional proof-of-concept implementation of the reporting system, built to test and demonstrate the research findings rather than to be production-ready.
STARR	Situation, Task, Action, Result, Reflection — a structured format commonly used to document competency evidence for HBO-i assessment.

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1. Context and Background

1.1 InnoBeweegLab and the Case

InnoBeweegLab designs active-friendly public spaces that encourage people to move, be active, and stay healthy, working through participation and co-creation, expert assessment, quantitative research, and data analysis. Its core output is evidence-based research reports: conclusions and recommendations that municipalities, urban planners, and health stakeholders use to shape neighbourhood and city development.

Producing those reports currently means manually combining three kinds of input — resident and stakeholder perspectives gathered through interviews and surveys, expert observations and assessments, and quantitative datasets — into a single, structured, evidence-grounded document. InnoBeweegLab has stated that the less time its team spends on this manual synthesis, the more time it can dedicate to the fieldwork and community engagement that are its actual mission. ([Reference first in interview](#))

This project plan concerns a group project in which two Master of Applied IT students each take on their own research trajectory within that shared case, working toward a single AI-supported system that ingests InnoBeweegLab's heterogeneous research data and drafts evidence-based reports, while keeping the researcher fully in control of interpretation and final recommendations.

2. Problem Statement

InnoBeweegLab's researchers must repeatedly transform large volumes of heterogeneous, unstructured research data — interview transcripts, survey responses, expert assessments, and quantitative datasets — into structured research reports whose conclusions and recommendations are traceably grounded in that data. This synthesis work is manual, slow, and does not scale with InnoBeweegLab's ambition to spend more time directly engaging communities and less time in the office producing documents.

The problem is not simply “generate a report from data with an LLM”. Two constraints make it substantially harder in practice:

- Evidence traceability and researcher control. A draft that reads well but is not clearly grounded in the underlying data is worse than no draft at all for an organisation whose entire value proposition is evidence-based recommendations. InnoBeweegLab has been explicit that the system must support researchers, not replace their interpretation, conclusions, or professional judgement.
- Requested a zero budget solution. InnoBeweegLab has no budget allocated for AI tooling, uses Microsoft office software, and works on standard office laptops (assume roughly 16GB RAM, no dedicated GPU). Any architecture that assumes a well-resourced cloud deployment or frontier-model access by default is not actually deployable for this organisation. There can be a possibility of using Azure AI Foundry if the costs are approved by the finance department.

Generalisation Beyond This Project

This problem is not unique to InnoBeweegLab. Any organisation that combines qualitative and quantitative research into recommendation-style reporting — policy institutes, consultancies, other community or health-focused NGOs, municipal research departments — faces the same synthesis bottleneck. Many of them share InnoBeweegLab's resource profile: limited or no AI budget, and staff working on ordinary business hardware rather than specialised infrastructure. - [REFERENCE](#)

Published research on closely comparable systems confirms this is a genuinely open problem rather than a solved one. A 2025 master's thesis at Chalmers University of Technology, built with Volvo Trucks, evaluated a multi-agent LLM system for automated engineering report generation from unstructured expedition logbooks — architecturally very close to this project's aim. Even using a well-resourced model mix (a 14B routing model, a 70B extraction model, and GPT-4o for writing), the resulting reports still scored far below human-written reports on factuality and usefulness in blind human evaluation, though the multi-agent architecture did measurably reduce hallucination compared to a naive single-prompt baseline, primarily through retrieval-augmented grounding.

The generalisable question this project addresses, then, is not whether AI can draft a report, but how much of the synthesis burden can responsibly be automated — under real budget and hardware constraints — while keeping evidence traceable and interpretive control with the human researcher.

3. Research Questions

This trajectory follows Fontys' DOT framework, which structures applied research around a central design question and five research types: Library, Field, Lab, Workshop, and Showroom. Each research type contributes its own sub-question; together they answer the central design question.

Central Design Question

CENTRAL DESIGN QUESTION

How can a provider-agnostic, multi-agent reporting architecture — deployable within InnoBeweegLab's zero-budget, standard-office-hardware constraints, and paired with a researcher-controlled validation step — be designed so that heterogeneous qualitative and quantitative research data is reliably transformed into traceable, evidence-based draft reports?

DOT Sub-Questions

Library — SQ1

What architectural patterns, retrieval strategies, and human-in-the-loop validation designs does existing literature identify as effective for AI-assisted report generation from heterogeneous data, and which of these remain viable under zero-budget, modest-hardware constraints?

Field — SQ2

What does InnoBeweegLab's current research-to-report workflow look like in practice, where are its main bottlenecks, and what would make researchers actually trust and adopt an AI-drafted report?

Lab — SQ3

Which combination of retrieval-augmented generation, model deployment strategy (locally-run small models versus a hybrid cloud fallback), and prompting technique produces the most grounded, least generic draft report sections when tested against InnoBeweegLab's own (anonymised) data and a standard 16GB-RAM office laptop?

Workshop — SQ4

How should the researcher-facing validation and review interface be co-designed with InnoBeweegLab researchers so that it supports trust, traceability, and efficient correction of AI-drafted content? (Primary evidence for Interaction, level 3.)

Showroom — SQ5

To what extent does the resulting prototype, evaluated by InnoBeweegLab stakeholders against human-written reports and a naive baseline, meet the traceability, factuality, and usefulness bar required for real adoption — and what generalisable design guidance follows for similarly resourced organisations? (Primary evidence for Software, level 4: producing generic knowledge, not just a working product.)

4. Applied Methods

The DOT framework triangulates evidence across five research types rather than relying on a single method. Each type below is mapped to the sub-question it answers and the concrete method used.

Type	Answers	Method
Library	SQ1	Structured literature review of multi-agent report generation, retrieval-augmented generation, and human-in-the-loop validation interfaces, including the Chalmers/Volvo thesis as a directly comparable precedent; review of InnoBeweegLab's existing report templates and past research documentation.
Field	SQ2	Semi-structured interviews and contextual inquiry with InnoBeweegLab researchers; workflow mapping of the current data-to-report process; document analysis of provided datasets, past reports, and reporting templates.
Lab	SQ3	Controlled technical experiments comparing architecture variants (naive single-prompt baseline vs. multi-agent with RAG) and model deployment tiers (a small model quantised to run on a 16GB office laptop vs. a hybrid call to a hosted API for the writing step), scored with an adapted faithfulness / factual-correctness / usefulness framework plus small-scale human review.
Workshop	SQ4	Participatory co-design sessions with InnoBeweegLab researchers to design the validation/review interface; iterative low-fidelity to functional prototyping; usability walkthroughs.
Showroom	SQ5	Structured evaluation session with InnoBeweegLab stakeholders comparing AI-drafted sections to human-written reports on real, anonymised data (Likert-scale + qualitative feedback); synthesis of findings into generalisable design guidance.

Technical Approach (Lab / Software)

The system under test is a multi-agent pipeline: separate agents (or agent roles) handle data ingestion, retrieval-augmented extraction from InnoBeweegLab's source material, section drafting, and assembly into a structured report, with a researcher validation step before anything is treated as final. The architecture is kept provider-agnostic — the model behind each agent role is swappable — specifically so the zero-budget, ordinary-hardware constraint can be tested empirically rather than assumed away: lighter, well-defined subtasks (extraction, tagging, retrieval) are candidates for a small model running locally, while the hardest step, final section synthesis, is where a hybrid fallback to a low-cost hosted model is evaluated as an explicit design alternative if local-only quality proves insufficient.

Evaluation Approach (Workshop / Showroom)

Evaluation combines an automated, RAGAS-inspired scoring approach (faithfulness to source data, factual correctness, and usefulness of the summary) with structured human evaluation by InnoBeweegLab researchers, mirroring the dual approach validated in comparable published research. Both the naive baseline and the multi-agent prototype are compared against a human-written reference on the same underlying data, so quality claims are relative and evidenced rather than assumed.

5. Project Deliverables

Per the Master of Applied IT assignment, this project combines research-oriented outputs with traditional IT products, so that the research findings are demonstrated through a working case study rather than only argued in writing.

Research-Oriented Outputs

- Background literature study, covering multi-agent report generation, RAG, HITL validation interfaces, and comparable published evaluations.
- Research paper, presenting the central design question, theory, method, findings, and generalisable design guidance.
- Research poster, summarising the project for the Showroom presentation.
- Evaluation report, documenting both the automated (RAGAS-inspired) and stakeholder-evaluation results, including the baseline vs. multi-agent vs. human-written comparison.

Traditional IT Products

- Requirements documentation: functional and non-functional requirements, user stories, and researcher personas, gathered during Field research.
- Architecture and design documentation: system architecture diagrams, data-flow diagrams, and the provider-agnostic abstraction-layer design (the core Software level 4 evidence).
- Interaction design documentation: the co-designed researcher validation/review interface, including the evaluative usability test (the core Interaction level 3 evidence).
- Test and evaluation documentation: the Lab-phase experiment design, scoring methodology, and results.
- Functional prototype: a proof-of-concept reporting pipeline (ingestion → retrieval-grounded extraction → draft generation → researcher validation) tested on real, anonymised InnoBeweegLab data, generalisable as a case study beyond InnoBeweegLab itself.
- Technical documentation, to support handover or future development by InnoBeweegLab or the group project team.

6. Stakeholder Overview

Stakeholder	Role	Interest in the Project
InnoBeweegLab research team	Primary user / data owner	Source of workflow knowledge, report templates, and (anonymised) research data; evaluates the prototype directly; ultimate adopter if the approach proves viable.
Municipalities, urban planners & health stakeholders	Indirect / downstream	Consume InnoBeweegLab's final reports; interested in faster turnaround without loss of report quality or evidence grounding.
Residents & community participants	Indirect data source	Original source of the qualitative input (interviews, surveys); interested in their voices being represented faithfully, not flattened into generic text.
Fontys supervisor / examiner	Academic quality assurance	Assesses whether the research process and competency evidence (Software L3→L4, Interaction L3) meet HBO-i graduation criteria.
Fellow group-project students	Peers, shared case	Work on their own competency trajectories within the same InnoBeweegLab case; coordination needed on shared data access and any shared infrastructure.
Me (project owner)	Researcher / developer	Responsible for answering the research questions, delivering the IT products, and producing valid competency evidence within the timeline below.

7. Scope

7.1 In Scope

- Research and design of a provider-agnostic, multi-agent reporting architecture (ingestion, retrieval-grounded extraction, draft generation, researcher validation).
- A functional prototype covering that pipeline, tested on real, anonymised InnoBeweegLab data, sized to what is realistically buildable within the timeline.
- Lab-phase technical experiments comparing a locally-run small model (targeting a 16GB-RAM office laptop) against a hybrid cloud-assisted configuration.
- A minimal but functional researcher validation/review interface, co-designed with InnoBeweegLab researchers and evaluated with a usability test.
- Evaluation of the prototype against a human-written reference and a naive baseline, using both automated and stakeholder-based methods.
- All research-oriented and IT deliverables listed in Section 5.

7.2 Out of Scope

- Production deployment of the system into InnoBeweegLab's actual operations.
- Full coverage of every data format or research method InnoBeweegLab has ever used — the prototype is scoped to the data made available during the project.
- Polished, production-grade UI/UX design; the validation interface is functional and demonstrable, not visually finished.
- Automating final conclusions or policy recommendations without a human sign-off step — by design, this stays out of scope permanently, not just for this project.
- Integration with municipality-side systems or any system outside InnoBeweegLab itself.
- Support for languages beyond English and Dutch as needed for the source data.

8. Risk Management

Risks are graded on likelihood and impact (Low / Medium / High), with a mitigation strategy defined for each. R1 and R3 are the highest-priority risks, being high on at least one dimension with direct consequences for whether the timeline in Section 9 holds.

ID	Risk Description	Likelihood	Impact	Mitigation Strategy
R1	Small, locally-run models on a 16GB-RAM office laptop are insufficient quality for grounded report synthesis.	Medium	High	Test this explicitly and early, in the Lab phase; design the architecture with a hybrid cloud-fallback option from the start rather than as a late patch.
R2	Limited or delayed access to real (anonymised) InnoBeweegLab data during early phases.	Medium	Medium	Secure data access during Field research in Month 1; prepare a small synthetic/stand-in dataset as a fallback for early Lab experiments.
R3	Time constraint: full scope not realistically finished before the Christmas break as planned.	Medium	High	MVP-first prioritisation, buffer time built into weeks 13–15, and strict adherence to the Out-of-Scope list in Section 7.2.
R4	Evaluation results are hard to interpret due to a small stakeholder sample and subjective judgement.	Medium	Medium	Combine structured Likert-scale human evaluation with an automated RAGAS-inspired scoring layer, following the dual-method approach validated in comparable published research.
R5	Coordination overhead with fellow group-project students sharing the same case and data.	Medium	Low	Keep sub-questions and deliverables scoped to my own trajectory; agree data-access and infrastructure boundaries early; regular sync with the group.

9. Approach and Planning

9.1 Methodology

The project follows the DOT framework's five research types as its overall structure (Section 3-4), executed through short, iterative two-week cycles similar in spirit to Scrum: each cycle ends with a concrete, reviewable output rather than only a status update. Regular check-ins with the Fontys supervisor and, where relevant, InnoBeweegLab stakeholders keep the direction aligned throughout, rather than only at phase boundaries.

9.2 Timeline Overview

The project runs from Monday 7 September 2026 to Saturday 30 January 2027 (19 working weeks plus a two-week winter break). All research, design, development, and stakeholder evaluation are completed before the break; January is reserved exclusively for analysing evaluation results and finalising the research report, poster, and presentation.

Phase	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15		16	17	18	19
	7 Sep	14 Sep	21 Sep	28 Sep	5 Oct	12 Oct	19 Oct	26 Oct	2 Nov	9 Nov	16 Nov	23 Nov	30 Nov	7 Dec	14 Dec	Break	4 Jan	11 Jan	18 Jan	25 Jan
1. Orientation & Problem Investigation				◆																
2. Design								◆												
3. Realisation (build prototype)													◆							
4. Stakeholder Evaluation															◆					
5. Analysis, Reporting & Finalisation																				◆

◆ marks the milestone at the end of each phase. The shaded “Break” column marks the two-week winter holiday (no scheduled project work).

9.3 Phase-by-Phase Breakdown

Phase 1 — Orientation & Problem Investigation (Weeks 1–4, 7 Sep – 2 Oct)

- Library + Field research: literature review, InnoBeweegLab workflow interviews, review of existing datasets and report templates.
- Draft and validate the central research question and DOT sub-questions with the supervisor.
- Milestone: research proposal, with confirmed research questions and data access, project plan, background research, Requirement Analysis delivered.

Phase 2 — Design (Weeks 5–8, 5 Oct – 30 Oct)

- Lab: architecture design (multi-agent pipeline, provider-agnostic abstraction layer), technical spike on 16GB-laptop model feasibility.
- Workshop: co-design sessions with InnoBeweegLab researchers for the validation/review interface.
- Milestone: architecture and interface design sign-off; technical results documented.

Phase 3 — Realisation (Weeks 9–13, 2 Nov – 4 Dec)

- Build the ingestion, retrieval-grounded extraction, and draft-generation pipeline.
- Build the researcher validation interface; integrate and iterate on real (anonymised) data.
- Milestone: demo-ready working prototype.

Phase 4 — Stakeholder Evaluation (Weeks 14–15, 7 Dec – 18 Dec)

- Send the prototype to InnoBeweegLab stakeholders for evaluation before the winter break.
- Run the structured evaluation alongside the automated RAGAS-inspired scoring.
- Milestone: prototype delivered and evaluation launched with InnoBeweegLab before Christmas.

HOLIDAY

Winter break: 19 December 2026 – 3 January 2027. No project work scheduled.

Phase 5 — Analysis, Reporting & Finalisation (Weeks 16–19, 4 Jan – 30 Jan)

- Analyse the stakeholder evaluation results and combine them with the automated scoring from Phase 3.
- Write the final research report and extract generalisable design guidance.
- Prepare and rehearse the research poster and Showroom presentation.
- Milestone: final report, poster, presentation, and competency evidence portfolio submitted by 20th January.

10. Quality Management and Success Criteria

10.1 Acceptance Criteria

- The prototype demonstrates the core pipeline: multi-source data ingestion, retrieval-grounded extraction, draft report generation, and a working researcher validation step.
- The architecture and its design decisions (including the local-vs-hybrid model choice) are backed by the Lab-phase experiments, not assumed.
- The prototype is evaluated on real, anonymised InnoBeweegLab data, using both automated and stakeholder-based methods, against a human-written reference and a naive baseline.
- All deliverables (requirements, design, evaluation, prototype, documentation) are complete and traceable back to the research questions in Section 3.
- InnoBeweegLab stakeholder feedback is incorporated into the interpretation of results and the final report.

10.2 Definition of Done

- All planned pipeline features are implemented and tested against the acceptance criteria above.
- The prototype is stable enough to demonstrate live in the Showroom session.
- Documentation (requirements, architecture, interaction design, evaluation) is complete and internally consistent.
- Evaluation results are documented with a measurable comparison across the baseline, the multi-agent prototype, and the human-written reference.
- The research report, poster, and presentation are finalised and the competency evidence portfolio (STARR-format, per Section “Open Items” below) is complete.

10.3 Quality Assurance Measures

- Iterative DOT cycles with a reviewable output at the end of each phase, rather than only at the very end of the project.
- Regular reviews with the Fontys supervisor and, at the Phase 4 milestone, InnoBeweegLab stakeholders.
- Version control and consistent documentation standards across all IT deliverables, to keep everything traceable.
- Evaluation methodology (automated + human) is itself modelled on a validated, published approach rather than invented ad hoc, to keep the results defensible.

10.4 Open Items to Verify with Supervisor / Examiner

- Whether the group project shares one central design question with individual sub-questions, or each student needs a fully independent central question.
- The exact expected format for competency evidence (portfolio, assessment interview, study plan, poster, etc.).

11. Work Breakdown Structure

The work breakdown structure decomposes the project into manageable activities across six categories, aligned with the phases in Section 9.

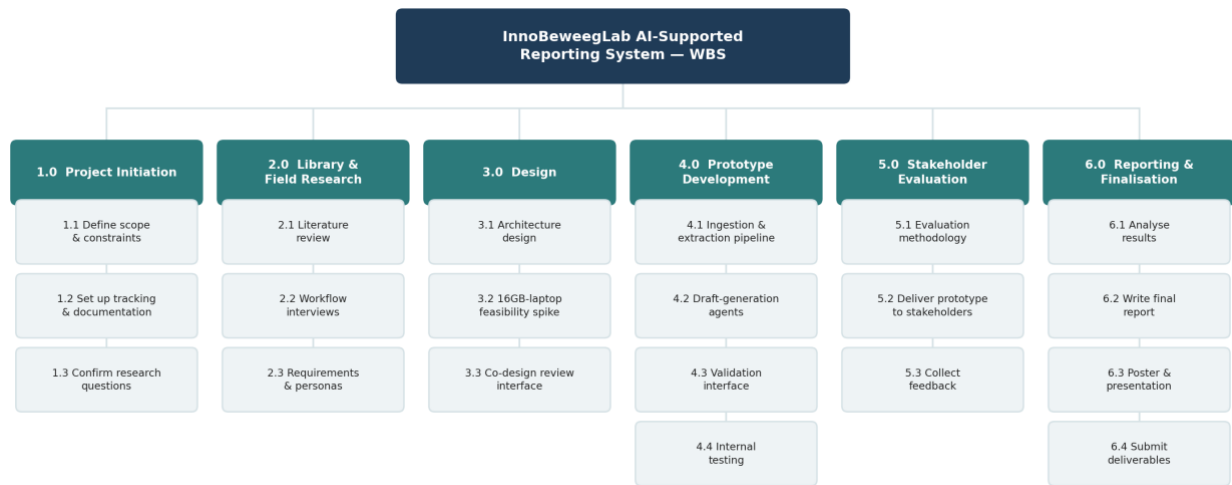


Figure 11.1: Work breakdown structure, showing the six top-level categories and their constituent activities.

1.0 Project Initiation

- 1.1 Define scope, stakeholders, and constraints (zero budget, 16GB-laptop hardware).
- 1.2 Set up project tracking and documentation structure.
- 1.3 Confirm the central design question and DOT sub-questions with the supervisor.

2.0 Library & Field Research

- 2.1 Literature review (MAS, RAG, HITL interfaces, comparable published evaluations).
- 2.2 InnoBeweegLab workflow interviews and document analysis.
- 2.3 Define personas, user stories, and functional/non-functional requirements.

3.0 Design

- 3.1 Provider-agnostic multi-agent architecture design.
- 3.2 Technical research: local small-model feasibility on 16GB RAM.
- 3.3 Co-design the researcher validation/review interface (Workshop).

4.0 Prototype Development

- 4.1 Build ingestion and retrieval-grounded extraction pipeline.
- 4.2 Build draft-generation agents and report assembly.
- 4.3 Build and integrate the researcher validation interface.
- 4.4 Internal testing and iteration on real anonymised data.

5.0 Stakeholder Evaluation

- 5.1 Prepare evaluation methodology (automated + Likert).
- 5.2 Deliver prototype to InnoBeweegLab stakeholders before the winter break.
- 5.3 Collect and log stakeholder feedback.

6.0 Reporting & Finalisation

- 6.1 Analyse evaluation results.

- 6.2 Write the final research report and design guidance.
- 6.3 Prepare research poster and presentation.
- 6.4 Submit final deliverables and competency evidence portfolio.

12. Sources

- InnoBeweegLab project brief: “Master of Applied IT Group project case: From data to evidence-based recommendations” (Sander Baartmans).
- HBO-i domain description, hbo-i.nl (to be verified against the exact edition used for assessment — see Section 10.4).
- DOT framework (Design & development-Oriented research Template/Triangulation), Fontys ICT / Saxion applied-research methodology.
- A. Nihlstrand & G. Lundberg, “Automatic Engineering Report Generation Using Multi-Agent Systems and Large Language Models”, master's thesis, Chalmers University of Technology & University of Gothenburg, with Volvo Trucks, 2025.
- RAGAS evaluation framework for retrieval-augmented generation.
- Earlier project work: GenAI provider cost comparison & business case, and free/local self-hosting feasibility research (this project's companion documents).

End of document.